# Enderton's Elements of Set Theory

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# Chapter 1

# Axioms and Operations

## 1.1 First Axioms

*Extensionality axiom*
If two sets have the exact same members, then they are equal:

$$\forall A \forall B[\forall x(x \in A \leftrightarrow x \in B) \rightarrow A = B]$$

(The nested implication is not an equivalence because the other direction is already true. Declaring it would be redundant.)

*Empty set axiom*
There is a set having no members:

$$\exists B \forall x(x \notin B)$$

*Pairing axiom*
For any sets $u$ and $v$, there is a set having as members just $u$ and $v$:

$$\forall u \forall v \exists B \forall x(x \in B \leftrightarrow x = u \text{ or } x = v)$$

*Union axiom, Preliminary form*
For any sets $a$ and $b$, there is a set whose members are those sets belonging either to $a$ or to $b$ (or both):

$$\forall a \forall b \exists B \forall x(x \in B \leftrightarrow x \in a \text{ or } x \in b)$$

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*Power set axiom*
For any set $a$, there is a set whose members are exactly the subsets of $a$:

$$\forall a \exists B \forall x (x \in B \leftrightarrow x \subseteq a)$$

We can rewrite $x \subseteq a$ in terms of the definition of $\subseteq$:

$$\forall t (t \in x \rightarrow t \in a)$$

This list will be further expanded later to include

- Subset axioms
- Replacement axioms
- Infinity axiom
- Regularity axiom
- Choice axiom

The union axiom will be also be restated in a stronger form.

### 1.1.1 Definitions

**Definition 1.1.** *$\emptyset$ is the set having no members.*

This definition bestows the name "$\emptyset$" on a certain set. When we write down such a definition we must be sure of two things: that there exists such a set and that there cannot be more than one set having no members. The empty set axiom provides the first fact and extensionality provides the second. Without both facts the symbol "$\emptyset$" would not be well defined.

**Definition 1.2.**

- *For any sets $u$ and $v$, the* ***pair set*** *$\{u, v\}$ is the set whose members are $u$ and $v$.*
- *For any sets $a$ and $b$, the* ***union*** *$a \cup b$ is the set whose members are those sets belonging either to $a$ or to $b$ (or both).*
- *For any set $a$, the* ***power set*** *$\mathscr{P}a$ is the set whose members are exactly the subsets of $a$.*

The existence and uniqueness proofs guaranteeing all of these definitions can be proved using the empty set axiom, the pairing axiom, the union axiom, and the power set axiom respectively, each in combination with the extensionality axiom.

We can use the pairing and union together to form other finite sets. Given any $x$ we can denote the pair set $\{x, x\}$ as the *singleton* $\{x\}$ (since the pair exists and is unique). For any given $x_1, x_2, x_3$ we can define

$$\{x_1, x_2, x_3\} = \{x_1, x_2\} \cup \{x_3\}$$

(since the union exists and is unique for any two particular sets). Similarly we can define $\{x_1, x_2, x_3, x_4\}$ and so forth.
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**Notation**

Our existence axioms have been of the form "there exists a set $B$ whose members are those sets $x$ satisfying the condition __"

$$\exists B \forall x (x \in B \leftrightarrow \_\_)$$

Where if some other sets $t_1, \ldots, t_k$ are involved in the condition, then the full version becomes

$$\forall t_1 \cdots \forall t_k \exists B \forall x (x \in B \leftrightarrow \_\_)$$

With the blank filled in be some expression involving $t_1, \ldots, t_k$ and $x$. The empty set axiom can be written as

$$\exists B \forall x (x \in B \leftrightarrow x \neq x)$$

Now let us be more general and consider any statement of the form

$$\forall t_1 \cdots \forall t_k \exists B \forall x (x \in B \leftrightarrow \_\_)$$

If the statement were true, then the set $B$, by the statement itself, as well as by equivalence, can be named by use of the abstraction notation:

$$B = \{x \mid \_\_\}$$

*My interpretation was that showing the existence and uniqueness of $B$ amounts to showing:

$$\forall t_1 \cdots \forall t_k [\exists B (\forall x (x \in B \leftrightarrow \_\_) \wedge \forall B' (\forall x (x \in B' \leftrightarrow \_\_) \rightarrow B' = B))]$$

The sets defined earlier can similarly be named in this notation:

$$\begin{aligned} \emptyset &= \{x \mid x \neq x\} \\ \{u, v\} &= \{x \mid x = u \text{ or } x = v\} \\ a \cup b &= \{x \mid x \in a \text{ or } x \in b\} \\ \mathscr{P} a &= \{x \mid x \subseteq a\} \end{aligned}$$

Note that it is not the case that all statements of the general form are true. For instance

$$\exists B \forall x (x \in B \leftrightarrow x = x)$$

is false as it asserts the existence of a set $B$ to which every set belongs.

### 1.1.2 More axioms